Training-free Video Temporal Grounding Enhanced by Contextual Region Cue based Test-time Adaptation

Anonymous submission

Abstract

Video Temporal Grounding (VTG) aims to localize events in untrimmed videos via natural language queries. While existing training-free methods using Visual-Language Models (VLMs) to achieve promising results by analyzing frame-text similarity curves, they suffer from two issues: (1) Fragmentation of continuous events due to irrelevant or low-quality frames within event period, and (2) Extension caused by hard negative frames out of event period that exhibit superficial visual similarity to the query but event doesn't actually occur. These problems stem from VLM methods' insufficient focus on fine-grained regional context, leaving the query's discriminative visual cues unable to assist localization. To address these challenges, we propose a training-free **Video-Level Adapter** that enhances grounding capability by performing test-time adaptation on vision-language models (VLMs) through contextual region-aware visual token analysis. The adapter consists of two main parts: (1) **Positive Cache**, which stores key visual tokens from discriminative relevant regions, providing contextual evidence to enhance similarity scores and reduce abrupt truncations from low-quality frames by maintaining temporal continuity; (2)**Outlier Cache**, which captures visual tokens including distracting regions with a superficial correlation to the query and alleviates the over-extension by reducing the impact of these regions. By jointly leveraging visual consistency and semantic discriminability, our method refine score curves and reduces localization errors. Experiments on Charades-STA, ActivityNet and QVHighlights show that our adapter significantly improves grounding accuracy without retraining.

Introduction

Video temporal grounding (VTG) focuses on identifying the segment that is most semantically relevant within untrimmed videos, based on open-ended natural language queries. Recent pre-trained models exhibit remarkable zero-shot grounding capabilities, which inspired the development of training-free zero-shot video temporal localization methods. These methods achieve favorable performance across datasets flexibly without time-consuming training by using Visual Language Models (VLMs) (Li et al. 2023; Radford et al. 2021) to compute semantic similarity score between each frame and query text and construct score curve. Analyzing the similarity score's temporal variation, they identify the query event's temporal segment.

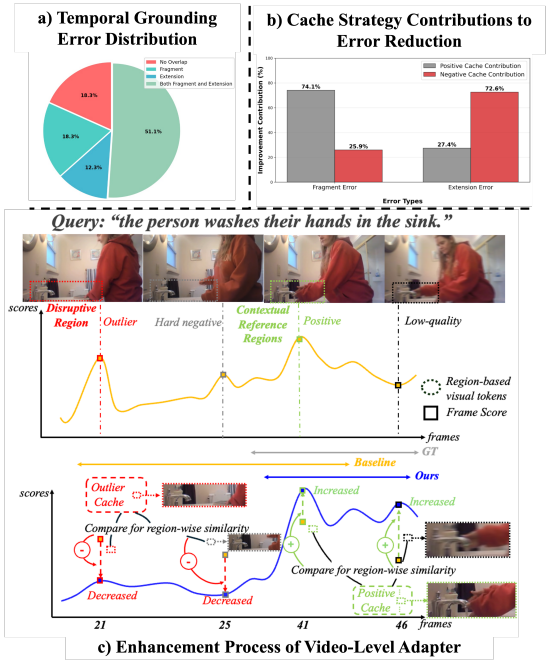


Figure 1: **Problem Formation**: Existing methods struggle with two issues: localization interval truncation from irrelevant/low-quality frames (e.g., motion blur, occlusion) and grounded segment extension from hard negatives. We avoid these with Positive and Outlier Caches, using region-based contextual visual similarity to enhance localization.

Zero-shot VLM-based methods (Luo et al. 2024; Xu et al. 2024a; Zheng et al. 2024) have shown impressive results but face key challenges. Relying on VLM's static pre-trained knowledge and scoring with frame-level visual-text comparison, they can ignore the video's internal contextual knowledge and lack attention to discriminative visual cues of local regions for the query event, leading to two common issues: *Fragmentation of Continuous Events* and *Extension by Hard Negatives* (as shown in Figure 1 (a)). Specifically, *1) Fragmentation of Continuous Events* is usually caused by methods incorrectly split events due to interference from low-quality or indirectly relevant frames (e.g., blur, occlusion).

In Figure 1 (c), frame 46 (a woman reaching for the tap) is blurry, causing a semantic mismatch that drops the score and truncates the segment prematurely, while the current methods ignore useful contextual visual cues (frame 41) of the clearer "washing hand" region. *2) Extension by Hard Negatives* can be caused by visually similar but irrelevant frames (hard negatives; e.g., frames 21, 25) mislead grounding by matching superficial cues (e.g., "hands near a sink" region) rather than distinctive ones (e.g., hand-water interaction region in frame 46), leading to erroneous segment extension. Mitigating this requires identifying and suppressing such local cues that mislead alignment with the text query.

To address such challenges, we propose a **Video-Level Adapter (VLA)**. During test-time, it adapts VLM to video and text inputs by caching region-discriminative visual tokens based on query-guided video context, then achieves region-wise visual comparisons between these tokens and each frame to aid grounding. Specifically, as shown in dashed-edge cropped regions in Figure 1 (c), we introduce visual tokens to capture fine-grained cues from query-relevant discriminative regions; these tokens are compared region-wise via similarity calculations with regions in each frame, enabling fine-grained query event localization. Stored in the adapter's two components—**Positive Cache** and **Outlier Cache**—they address the aforementioned issues. To mitigate *Fragmentation of Continuous Events*, we boost low-quality frame scores (black-dashed rectangles on score curves) using reference visual region cues in **Positive Cache**, which stores tokens focusing on regions most relevant to the query event (serving as visual references). When interfering frames share similar cues (green- and black-dashed cropped regions in frames 41 and 46, both showing hands touching flowing tap water), this similarity enhances their scores (frame 46's score shifts from yellow to blue curve), preventing sudden drops and avoiding fragmentation. Similarly, to mitigate *Extension by Hard Negatives*, we suppress their scores using distracting superficial cues in **Outlier Cache** (gray-dashed rectangles); these cues come from "outlier frames"—which show high query-semantic similarity but lack temporal coherence (unlikely to contain the query event, e.g., anomalous frames far from other high-scoring ones like frame 21)—with high similarity often stemming from superficially relevant but incorrect regions (red-dashed cropped region showing a sink and hand without "washing"). Comparing with these cues and subtracting their impact on query scores (frame 25 score variation) avoids extension caused by hard negative frames.

Our VLA operates test-time adaptation for VTG task through two mechanisms. First, it caches visual tokens from the test video and treats them as video-specific parameters, injecting the video's own in-domain visual knowledge into the VLM without training. Second, VLA stores the most discriminative region tokens as context reference and reweights each frame's score by region-wise comparison with cashed tokens, enabling the VLM trained from image-text pairs to link consecutive events while suppressing redundant region similarities across adjacent frames.

Figure 1 (b) shows that the reduction in fragment errors achieved by the adapter primarily stems from the positive cache (74.1%), while the negative cache is more effective in mitigating extension errors (72.6%). Thanks to the adapter, our method achieves a new zero-shot state-of-the-art (SOTA) on both Charades-STA (Gao et al. 2017) and ActivityNet (Krishna et al. 2017). Since the Adapter automatically extracts query-relevant context as video-domain knowledge , making it particularly valuable for out-of-distribution (OOD) settings. Experiments under OOD also yield SOTA results, confirming its strong generalizability.

To sum up, our contributions are as follows: (1) We design Video-Level Adapter: a training-free module to enhance VLM-based zero-shot video temporal grounding by using contextual visual cues based on local regions, stored in Positive and Outlier Caches, to improve event understanding for grounding through fine-grained regional comparisons during test-time without parameter updating. (2) The Positive Cache improves the scores of low quality frames within an event with contextual reference regions, helping to mitigate event fragmentation, while the Outlier Cache prevents over-extension by reducing the influence of distracting visual patterns and suppressing hard negative frames. (3) Our method achieves state-of-the-art results on the Charades-STA (Gao et al. 2017), ActivityNet (Krishna et al. 2017) and QVHighlights datasets under various setting comparing with current zero-shot method, showing the effectiveness and generalization capability of our Video-Level Adapter.

Related Works

Video Temporal Localization

Video temporal grounding (VTG) aims to localize video segments matching textual queries, with approaches categorized into fully supervised, weakly supervised, and training-free methods. Fully supervised methods (e.g., (Zhang et al. 2020; Gao et al. 2017; Huang et al. 2022)) rely on manual timestamp annotations, suffering from annotation bias and limited generalization. Weakly supervised methods (e.g., (Zheng et al. 2022b; Nam et al. 2021; Zheng et al. 2023)) use incomplete annotations or pseudo-labels but remain constrained by pre-training data limitations. In contrast, training-free VTG methods leverage visual-linguistic models (VLMs) or large language models (LLMs) to exploit rich web-data knowledge, achieving strong performance. For instance, (Luo et al. 2024; Xu et al. 2024a) measure semantic similarity between VLM-generated video captions and queries, while TFVTG (Zheng et al. 2024) uses LLMs to decompose queries and analyze VLM similarity curves. However, these methods underutilize intrinsic connections among video visual features. Our work combines VLM pre-trained knowledge with adapters that introduce fine-grained visual features from input videos, providing in-domain visual references across frames to enhance training-free VTG.

Test-time adaptation

Test-time adaptation is an effective training-free strategy for deploying pre-trained vision-language models (VLMs) in downstream tasks (e.g., image classification, text-image retrieval), offering low computational cost and addressing distribution shifts. Examples include MEMO (Zhang, Levine,

and Finn 2022) (ensuring prediction consistency across augmented test samples), TPT (Shu et al. 2022) (dynamic prompt optimization), DiffTPT (Feng et al. 2023) (diversified augmentations), Tip-Adapter (Zhang et al. 2022) (few-shot caching without parameter updates), and TDA (Karmanov et al. 2024) (mitigating pseudo-label noise). Notably, video temporal grounding (VTG) faces unique challenges due to open-ended natural language queries—each video is paired with distinct queries, hindering cross-video visual information reuse without training. To address this, we propose exploiting temporal continuity within a single video by extracting visual region cues from multiple frames to aid query adaptation. Our method is the first to leverage fine-grained similarities between cross-frame visual region tokens in the same video for test-time adaptation in VTG. It efficiently improves the performance of VLMs on this task without requiring any parameter updates.

Methodology

Baseline Revisit

Given a video containing N frames and a query text, Training-Free Video Temporal Grounding (TFVTG) (Zheng et al. 2024) leverage scoring-based grounding method to localize the start and the end point of the query event. Specifically, TFVTG first extracts visual features $F \in \mathbb{R}^{N\times D}$ of video frames and textual query embeddings $f_q \in \mathbb{R}^D$, where D is the feature dimension. It calculate the cosine similarity between each frame and the query text to obtain a score curve $S_I \in \mathbb{R}^N$. A set of segments $W = [W_1, W_2, \ldots, W_n]$ with varying lengths and start/end points serves as proposals. For a proposal segment W_p starting at frame i and ending at frame j, its event score is computed as the sum of a dynamic term S^{dynamic} and a static term S^{static}. The dynamic term tracks significant similarity increases to mark the event start, while the static term compares the segment's average similarity with that of other frames to check overall relevance. Concretely, the score for W_p is calculated as $S^{final}_{i,j} = \max_{k=i}^{j}\left(S^{dynamic}_{i,k} + S^{static}_{k,j}\right)$, with each term's subscripts indicating their calculation intervals. Based on this scoring mechanism, the Top-K proposals in W based on S^{final} serve as the final predictions. Though achieved promised performance, they overlook video-internal context and fail to attend to regional visual cues for the query event, relying solely on VLM's static pre-trained knowledge pre-trained from image-text pair for grounding.

Overview

Figure 2 shows an overview of our method. To leverage discriminative region-level visual cues to aid grounding, we propose a Video-Level Adapter based on TFVTG. Such cues are selected from region-wise frame tokens $F_v \in \mathbb{R}^{N\times Q\times D}$ extracted by BLIP-2 Q-former(Li et al. 2023), where Q is the number of visual tokens for each frame and different tokens can attend to different regions of the frame. Furthermore, to mitigate erroneous fragmentation and extension, the adapter select tokens for Positive Cache and Outlier Cache respectively. The former one increases low-quality frames' scores with visual cues most relevant to the query text to prevent fragmentation, and the latter one suppresses hard negative frames' scores with superficial relevant cues of outlier high value frames to prevent extension. Specifically, we calculate the positive score S_P and the negative score S_N based on the similarity of F_v and the tokens in the Positive and Outlier cache. The weighted sum S_F of S_I, S_P, and S_N serves as the refined score after adaptation of the test time to the contextual regions of the input video and is used to locate the final grounding result following TFVTG.

Video-Level Adapter

Positive Cache. To prevent erroneous fragmentation caused by interfering frames, we introduce a positive cache that stores the visual tokens most relevant to the query as contextual references, leveraging the intrinsic visual continuity of the shared discriminative visual cues within the query event to increase the score of these frames. We pick the top k_p tokens from F_v that have the highest cosine similarity scores with f_q and store them in the positive cache as contextual reference as they are more likely to focus on the region with discriminative visual cues, denoted as $C_p \in \mathbb{R}^{k_p\times D}$. We select multiple tokens to capture query-relevant visual cues across spatiotemporal dimensions. These tokens can be complementary video in-domain parameters for the VLM, providing context as references and reflecting relevance to the query event through region-wise visual comparison. The region-wise comparison is achieved by calculating similarity $S_p \in \mathbb{R}^N$ between the cached visual tokens and F_v following Tip-Adapter (Zhang et al. 2022):

$$S_P = \max_q \sum_{i=1}^{k_p} \frac{\alpha \exp\left(-\beta\left(1 - F_v C_p^T\right)\right)_{n,q,i}}{k_p} \in \mathbb{R}^N \quad (1)$$

where α and β stands for a modulating hyper-parameter and $\max(\cdot)$ selects the max number in the second dimension. Such computing process mimics a multi-layer perceptron and incorporates video in-domain features as part of the model parameters to assist VTG during test-time.

Algorithm 1: Constructing Outlier Cache

1: **Input:** Tokens P with the highest relevance to the query in each frame, Maximum cluster size k, Value threshold τ, Initial score cure S_I
2: $P_{\text{high}} \leftarrow \{p \in P \mid S_I(p)$ is in the top $\tau\%\}$
3: Apply DBSCAN (Ester et al. 1996) on P_{high} to obtain clusters based on the tokens' frame indexes
4: Merge clusters based on distance
5: Identify the largest cluster $\mathbf{C}_{\text{max}}$
6: $\mathbf{O} \leftarrow P_{\text{high}} \setminus \mathbf{C}_{\text{max}}$
7: Construct Outlier Cache $C_n \leftarrow$ top k_n points in O

Outlier Cache. To mitigate event boundary extension induced by hard negative frames, we introduce an outlier cache to identify the most distracting regions outside the event period, which serve as "anchors" to suppress scores of corresponding frames. Since a key challenge is that the

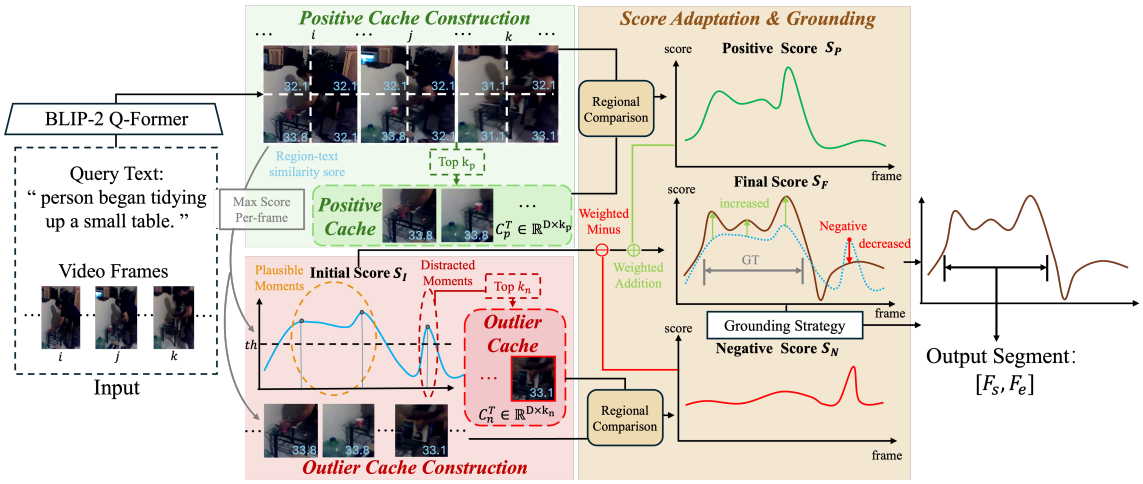


Figure 2: **Overview of our proposed methods.** Given the input video and query, visual tokens are extracted from the video using BLIP-2 Q-former, with an initial score S_I calculated for each frame. These tokens are categorized into Positive and Outlier Caches by relevance, yielding a positive score S_P and negative score S_N. The final score S_F is a weighted combination of S_I, S_P, and S_N, and temporal grounding is applied to S_F to generate the output segment.

specific event period is unknown, we try to leverage temporal coherence to locate such distracting regions.

The construction process of the Outlier Cache is shown in Algorithm 1. Specifically, to select the most misleading regions of each frame, we extract candidates from the high-value set P_{high}—defined as the top $\tau\%$ scoring tokens within $P \in \mathbb{R}^{N\times D}$, where P consists of the most query-related tokens in each frame. Then, to locate the distracting regions based on temporal coherence, we utilize the DBSCAN (Ester et al. 1996) algorithm to cluster these high-value points based on their frame indices, identifying points that are temporally close to each other. We merge clusters according to the distances between them to achieve the integration of information. The temporal segment of the largest cluster is regarded as the plausible moments when the query event occurs, while the high-value areas outside it are viewed as moments of distraction. we define high-value points not included in the largest cluster are identified as outlier points. These frames—with high relevance to the query text embedding yet low temporal continuity—are more likely to be distracting due to accidental overfitting on visually superficial event correlations. Such frames' token can be reference negative samples for score calibration of hard negative frames. We select the top k_n tokens of outlier high-value points and store them in the Outlier Cache, denoted as C_n. Similar to S_P, we define the negative score $S_N \in \mathbb{R}^N$ as :

$$S_N = \sum_{i=1}^{k_p} \frac{\alpha \exp\left(-\beta\left(1 - F_v C_p^T\right)\right)_{n,i}}{k_p} \in \mathbb{R}^N \qquad (2)$$

Final Localization. The final score is calculated by adding the product of the positive weight and the positive cache score, then subtracting the product of the negative weight and the negative score from the overall score:

$$S_F = S - w_n S_N + w_p S_P \qquad (3)$$

Where w_N and w_P are the hyper-parameters for the weights of S_P and S_N. Then, we follow the method of the TFVTG (Zheng et al. 2024)'s VLM localizer to generate the final grounding results based on score curve S_F. In this way, our adapter acts as a lightweight test-time adaptation step: it keeps the backbone frozen and recalibrates the output scores on-the-fly, achieving distribution alignment for the current video without any gradient updates.

Experiments

Experimental Setup

Datasets. We evaluate our method on two established datasets, ActivityNet Captions (Krishna et al. 2017) and Charades-STA (Gao et al. 2017). The *ActivityNet Captions* dataset comprises 20,000 videos, collected specifically for the task of video captioning. This dataset includes 37,417/17,505/17,031 video-query pairs in the train, val_1, and val_2 splits, respectively. Following prior studies (Luo et al. 2024; Wang et al. 2022b), we report performance metrics on the val_2 split. The *Charades-STA* dataset is constructed from the Charades dataset, containing 12,408/3,720 video-query pairs in the train/test splits. Furthermore, we present our findings based on the test split performance. We benchmark our method on QVHighlights (10,148 YouTube videos), using its 15% validation split following (Lei, Berg, and Bansal 2021). We test our method against existing approaches within IID, OOD, and cross-dataset frameworks following TFVTG(Zheng et al. 2024), aiming to highlight the generalization capabilities of our approach.

Evaluation Metrics. We adopt the evaluation metrics 'R@m' and 'mIoU' following previous research (Luo et al. 2024; Wang et al. 2022b), where 'm' represents a predefined temporal Intersection over Union (IoU) threshold. The 'R@m' metric quantifies the proportion of predictions with an IoU exceeding 'm', while 'mIoU' represents the average

Method	Setting	Charades-STA				ActivityNet Captions			
		R@0.3	R@0.5	R@0.7	mIoU	R@0.3	R@0.5	R@0.7	mIoU
2D-TAN (Zhang et al. 2020)	fully	-	39.81	23.25	-	58.75	44.05	27.38	-
EMB (Huang et al. 2022)		**72.50**	58.33	39.25	**53.09**	**64.13**	44.81	26.07	**45.59**
EaTR (Jang et al. 2023a)		-	**68.47**	**44.92**	-	-	**58.18**	**37.64**	-
CPL (Zheng et al. 2022b)		66.40	49.24	22.39	-	55.73	31.37	-	-
Huang et al. (Huang, Yang, and Sato 2023)		**69.16**	**52.18**	**23.94**	**45.20**	**58.07**	**36.91**	-	**41.02**
Kim et al. (Kim et al. 2023)	unsup. [1]	52.95	37.24	19.33	36.05	47.61	**32.59**	15.42	31.85
SPL (Zheng et al. 2023)		**60.73**	**40.70**	**19.62**	**40.47**	**50.24**	27.24	15.03	**35.44**
GroundingGPT (Li et al. 2024)	fully [2]	-	29.6	11.9	-	-	-	-	-
VTimeLLM-13B (Huang et al. 2023)		**55.3**	**34.3**	**14.7**	**34.6**	**44.8**	**29.5**	**14.2**	**31.4**
Momentor-7B (Qian et al. 2024)	zero-shot	42.6	26.6	11.6	28.5	42.9	23.9	12.4	29.3
Luo et al. (Luo et al. 2024)		56.77	42.93	20.13	37.92	48.28	27.90	11.57	32.37
VTG-GPT (Xu et al. 2024a)		59.48	43.68	**25.94**	39.81	47.13	28.25	12.84	30.49
TRACE (Guo et al. 2024)		63.7	43.7	21.0	41.5	-	-	-	-
Timesuite (Zeng et al. 2024)		**69.9**	48.7	24.0	-	-	-	-	-
TFVTG (Zheng et al. 2024) w/o LLM		65.46	48.01	22.07	43.37	48.84	26.64	13.10	33.61
Ours w/o LLM		67.68	51.53	25.45	45.30	49.25	27.10	13.34	33.95
TFVTG (Zheng et al. 2024) w LLM		67.04	49.97	24.32	44.51	49.34	27.02	13.39	34.10
Ours w LLM		68.27	**52.04**	25.43	**45.70**	**49.53**	**27.34**	**13.55**	**34.32**

Table 1: Evaluation Results on the Charades-STA and ActivityNet Captions Datasets.

Method	Setting	R1		mAP	
		@0.5	@0.7	@0.5	avg
MH-DETR(Xu et al. 2024b)	fully	60.84	44.90	60.76	39.26
EaTR(Jang et al. 2023b)		61.36	45.79	61.86	**41.74**
MS-DETR(Ma et al. 2025)		**66.9**	**51.68**	**67.19**	**46.00**
VTG-GPT (Xu et al. 2024a)	zero-shot	54.26	38.45	54.17	30.91
TFVTG (Zheng et al. 2024)		65.03	40.65	62.14	33.50
Ours		**65.48**	**41.35**	**62.5**	**33.78**

Table 2: Performance comparison on the QVHighlights.

IoU across all predictions.

Implementation Details. The BLIP-2 Q-former (Li et al. 2023) is utilized as VLM localizer. We downsample the input video to 3 FPS and calculate the similarity between video frames and text following TFVTG(Zheng et al. 2024). For the best hyperparameters on Charades-STA (Gao et al. 2017) dataset, our settings are as follows: the positive cache weight w_{p} and outlier cache weight w_{n} are 0.6 and 0.4 respectively, the shot capacity of positive cache k_{p} and negative cache k_{n} are 4 and 1 respectively, The hyperparameters reused in TFVTG (Zheng et al. 2024) remain the same.

Comparison with the SOTA methods

Comparison under the IID setting. We evaluate our approach on three public benchmarks—Charades-STA, ActivityNet-Captions, and QVHighlights—using the same score-based grounding framework as TFVTG (Zheng et al. 2024), both *with* and *without* TFVTG's LLM strategy. As reported in Table 1, our zero-shot model delivers the strongest results on Charades-STA and ActivityNet-Captions even *without* LLM support: it surpasses TFVTG by **3.52%** (R@0.5) and TRACE (Zeng et al. 2024) by **2.83%** (R@0.3) on Charades-STA, and exceeds the finetuned LLM methods like VTimeLLM-13B (Huang et al. 2023) on ActivityNet. Our method slightly underperforms TimeSuite (Zeng et al. 2024) on the R@0.3 metric, this is because it fine-tunes LLMs using additional collected data. In contrast, ours requires no training at all and performs better on other metrics. As shown in Tables 1 and 2, when scenes grow more complex, e.g., multi-target or densely annotated videos, our adapter-cached context cues enable richer query understanding, the Adapter thus outperform TFVTG on ActivityNet and QVHighlights. Unlike prior works like VideoLLaMA (Zhang, Li, and Bing 2023) and VTime-LLM (Huang et al. 2023), which rely on costly LLM fine-tuning or VQA pipelines, our method improves grounding purely through VLM-generated visual token semantics, incurring no extra training overhead. Besides, when we optionally add the same LLM reasoning module used by TFVTG, performance increases further, confirming that our cache-based refinement is also complementary to LLM-driven reasoning and can flexibly integrate with it. Our superiority over baseline TFVTG arises from mitigating its frame-level comparison limitation: our cached tokens enables finer contextual region-aware understanding, which is critical for grounding, especially when consecutive frames are visually similar.

Comparison under the OOD setting. By acquiring query-relevant visual cues according to query text from each video as in-domain priors, we expect our method to generalize well in out-of-distribution (OOD) scenarios; therefore, experiments in three OOD settings are conducted: novel location and novel text and cross-dataset. For novel location in Table 3, following DCM (Yang et al. 2021), we inserted a random generated video segment at the test video's beginning. Additionally, on the Charades-CD (Yuan et al. 2021) dataset in Table 4, which alters the distribution of target location by resplitting the training and testing sets, our method outperformed recent fully-supervised methods without training. For novel text, we tested on Charades-CG (Li et al. 2022) in Table 5 like VISA (Li et al. 2022). 'novel-composition' indicates the text contains novel combinations of training vocabulary, while 'novel-word' indicates the text contains novel words. Under novel-word, our method is the best. In novel-composition setting, though not surpassing methods specifically fine-tuned under fully supervised sce-

Method	Setting	Charades-STA						ActivityNet-Captions					
		OOD-1			OOD-2			OOD-1			OOD-2		
		R@0.5	R@0.7	mIoU	R@0.5	R@0.7	mIoU	R@0.5	R@0.7	mIoU	R@0.5	R@0.7	mIoU
LGI (Mun, Cho, and Han 2020)	fully	42.1	18.6	41.2	35.8	13.5	37.1	16.3	6.2	22.2	11.0	3.9	17.3
MMN (Wang et al. 2022b)		31.6	13.4	33.4	27.0	9.3	30.3	20.3	7.1	26.2	14.1	**5.2**	20.6
VDI (Luo et al. 2023)		25.9	11.9	26.7	20.8	8.7	22.0	**20.9**	7.1	**27.6**	**14.3**	**5.2**	**23.7**
DCM (Yang et al. 2021)		**44.4**	**19.7**	**42.3**	**38.5**	**15.4**	**39.0**	18.2	**7.9**	24.4	12.9	4.8	20.7
CNM (Zheng et al. 2022a)	weakly	9.9	1.7	21.6	6.1	0.5	16.6	**6.1**	**0.4**	21.0	**2.5**	0.1	16.8
CPL (Zheng et al. 2022b)		**29.9**	**8.5**	**32.2**	**24.9**	**6.3**	**30.5**	4.7	**0.4**	**21.1**	2.1	**0.2**	**17.7**
PSVL (Nam et al. 2021)	unsup.	**3.0**	0.7	8.2	**2.2**	0.4	6.8	-	-	-	-	-	-
PZVMR (Wang et al. 2022a)		-	**8.6**	**25.1**	-	**6.5**	**28.5**	-	**4.4**	**28.3**	-	**2.6**	**19.1**
Luo et al. (Luo et al. 2024)	zero-shot	40.3	18.2	38.2	38.9	17.0	37.8	18.4	6.8	21.1	18.6	7.4	20.6
TFVTG (Zheng et al. 2024)		45.9	20.8	43.0	43.8	20.0	42.6	20.4	**11.2**	31.7	18.5	10.0	30.3
Ours		**48.28**	**22.34**	**44.11**	**47.15**	**22.23**	**44.00**	**21.16**	11.1	**31.95**	**19.3**	**11.2**	**31.4**

Table 3: Results under OOD setting on the Charades and ActivityNet Dataset.

Method	Setting	Charades-CD		
		R@0.3	R@0.5	R@0.7
2D-TAN (Zhang et al. 2020)	fully	43.45	30.77	11.75
TSP-PRL (Wu et al. 2020)		31.93	19.37	6.20
SCDM (Yuan et al. 2019)		**52.38**	**41.60**	**22.22**
WSSL (Duan et al. 2018)	weakly	**35.86**	**23.67**	**8.27**
SPL (Zheng et al. 2023)	unsup.	62.96	38.25	15.53
TFVTG (Zheng et al. 2024)	zero-shot	65.07	**49.24**	23.05
Ours		**65.84**	49.09	**23.82**

Table 4: OOD setting results on the Charades-CD Dataset.

Method	Setting	novel-composition		novel-word	
		R@0.5	mIoU	R@0.5	mIoU
2D-TAN (Zhang et al. 2020)	fully	30.91	29.75	29.36	28.47
VISA (Li et al. 2022)		45.41	**42.03**	**42.35**	**40.18**
DeCo (Yang et al. 2023)		**47.39**	40.70	-	-
WSSL (Duan et al. 2018)	weakly	3.61	8.26	2.79	**7.92**
CPL (Zheng et al. 2022b)		**39.11**	**35.53**	**45.90**	**22.88**
Luo et al. (Luo et al. 2024)	zero-shot	40.27	-	45.04	-
TFVTG (Zheng et al. 2024)		43.80	40.19	56.26	46.90
Ours		**44.80**	**41.15**	**56.98**	**47.37**

Table 5: OOD setting results on the Charades-CG Dataset.

nario like VISA (Li et al. 2022) and DeCo (Yang et al. 2023), it still outperformed other fully-supervised methods and TFVTG (Zheng et al. 2024). Besides, for the cross-dataset setting (Bao and Mu 2022) in Table 6, our method also outperforms TFVTG (Zheng et al. 2024).These results demonstrate that our adapter exhibits strong generalization.

Ablation Studies

Effectiveness of each component. Table 7 shows the ablation studies on the proposed modules. (1) As shown in the 2nd row, the implementation of the positive cache improves upon the baseline by 2.58% in the R@0.7 metric and by 2.55% in the R@0.5 metric. (2) As shown in the 3rd row, the implementation of outlier cache further leads to consistent performance improvement, increasing the mIoU metric by 0.46% and the R@0.5 metric by 0.97% compared with method only using the positive cache.

Adapter's performance with different hyperparameter.

Method	R@1 R@0.5	R@1 R@0.7	R@5 R@0.5	R@5 R@0.7
2D-TAN (Zhang et al. 2020)	15.81	6.30	59.06	31.53
Debias-TLL (Bao and Mu 2022)	21.45	10.38	62.34	32.90
TFVTG (Zheng et al. 2024)	49.97	24.32	83.5	42.2
Ours	**52.04**	**25.43**	**84.9**	**43.7**

Table 6: Cross-dataset performance when training on ActivityNet captions and evaluate on Charades-STA.

positive cache	outlier cache	R@0.5	R@0.7	mIoU
×	×	48.01	22.07	43.37
✓	×	50.56	24.65	44.84
✓	✓	**51.53**	**25.45**	**45.30**

Table 7: Ablations on positive and outlier cache.

The model performance variations with hyperparameters W_{pos}, W_{neg}, k_{p}, and k_{n} are shown in the Figure 3. Compared with TFVTG (orange line), the model achieves obvious improvements under different hyperparameter settings, indicating that hyperparameters have little impact and thus verifying the robustness of the adapter.

Outlier cache construction strategy. We systematically tested other outlier cache construction strategies (Table 8). The baseline uses DBSCAN clustering based on frame indices to select high-scoring visual tokens outside the largest cluster, corresponding to Algorithm 1 line 5. We also tried random selection (row 1) and enhancing clustering robustness via inter-frame dissimilarity matrices following (Nam et al. 2021) (row 2), both leading to performance drops—indicating our method better identifies temporal outlier high-value points with minimal hyperparameters. Additional experiments with naive frame index continuity clustering and GMM (Reynolds et al. 2009) showed clustering method choice has minimal impact on metrics, demonstrating the outlier cache's robustness to such strategies.

Robustness to disturbances. To validate robustness, we tested Charades-STA data with random Gaussian blur/occlusion. Figure 4 shows that our method degrades less than TFVTG under disturbances, confirming that the introduction of contextual cues in the positive cache and visually comparison enhances the robustness to such interference.

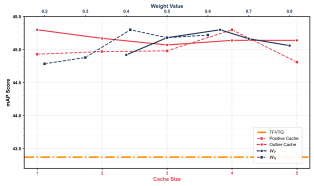


Figure 3: Performance variation with shots and weight.

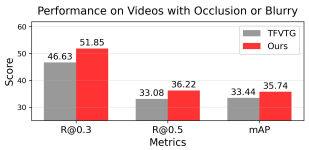


Figure 4: Results on videos with occlusion or blurry.

Selection Strategy	R@0.3	R@0.5	R@0.7	mIoU
Random Pick	66.57	50.15	24.31	44.49
Visual Similarity	66.97	50.36	24.51	44.71
Frame only(naive clustering)	67.34	51.45	25.54	45.07
Frame only(GMM)	67.39	51.53	25.03	45.13
Frame only (DBSCAN)	**67.68**	51.53	25.45	**45.30**

Table 8: Comparison with different outlier cache construction strategies.

Region-level cue analysis. To validate the importance of discriminative regional cues, we tested adapter's performance when caching global frame-level features by averaging all token features of the top-k frames most similar to the query. As in Table 9 row 2, it performs worse than our regional cue-based scheme, due to global features diluting salient regional signals and introducing background noise.

Video-Level Adapter in different score-based temporal grounding methods To validate our adapter's generalization, we integrated it with another *zero-shot* temporal grounding method VTG-GPT (Xu et al. 2024a)on the Charades-STA dataset, as shown in Table 10. Our method achieved a 3.7% mIoU improvement compared to the results we reproduced, showing the generalizability of the adapter.

Video-Level Adapter enables better and more efficient utilization of LLMs.. We find the adapter introduces a new method to leverage LLMs' query decomposition capability by caching both sub-query and original query tokens as contextual information. Unlike TFVTG, which requires repeated inference for each subquery, such a method leverages multiple subquery information and requires only a single inference. The result (Table 11 row 2, LLM Subquery Cache) shows it not only improves performance but also reduces processing time by 33% (from 15 to 10 minutes on Charades-STA), without significantly increasing memory usage, further proving the efficiency of the adapter.

Method	R@0.5	R@0.3	Memory	Time Cost
TFVTG w LLM	49.97	24.32	4467MB	15 min
LLM Sub-query Cache	51.77	25.43	4572MB	10 min

Table 11: Video-Level Adapter enables better and more efficient utilization of LLMs.

Qualitative Results

Figure 5 shows a qualitative visualization of a real sample. The positive cache selects frames 38 and 48 (green dashed lines), which show high activation in key regions

Method	R@0.5	R@0.7	mAP
Ours	**51.53**	**25.45**	**45.30**
global cue	50.48	24.38	44.67

Table 9: Comparison with global cue based cache features..

Methods	R@0.3	R@0.5	R@0.7	mIoU
TFVTG	64.46	48.01	22.07	43.37
+ VL Adapter	**67.68**	**51.53**	**25.45**	**45.30**
VTG-GPT	53.60	10.04	1.57	32.40
+ VL Adapter	**57.42**	**15.29**	**4.42**	**34.32**

Table 10: Video-Level Adapter with different baseline.

(hand and table) and boost scores for visually similar neighboring frames (e.g., frame 44), enhancing temporal continuity. The outlier cache collects frame 62 (red dashed line), whose tokens contain distracting semantics (e.g., hand moving a box), effectively suppressing scores of nearby frames and reducing interference. Together, these two caches enable more precise localization.

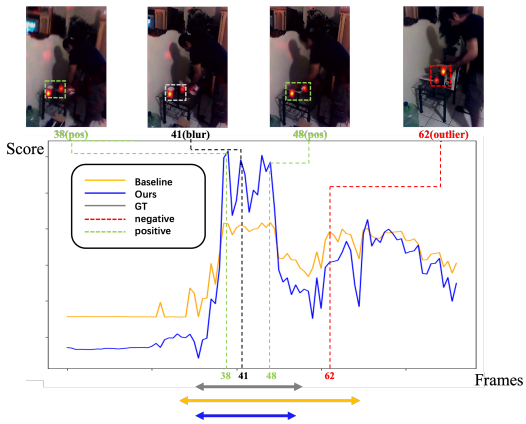


Figure 5: Qualitative Result.

Conclusion

We presents a novel training-free video-level adapter to enhance video temporal grounding by importing in-domain context and region-wise visual cues. It integrates two key components: a Positive Cache aggregating multi-frame semantics to boost event representation, and an Outlier Cache suppressing visually similar but irrelevant frames. Using contextual fine-grained visual tokens based score enhancement strategy, our method achieves state-of-the-art zero-shot video temporal grounding without backpropagation. Experiments across IID, OOD, and other settings on Charades-STA, ActivityNet, and QVhighlights confirm its superior performance, validating effectiveness and practical potential for training-free video temporal grounding.

References

Bao, P.; and Mu, Y. 2022. Learning sample importance for cross-scenario video temporal grounding. *arXiv preprint arXiv:2201.02848*.

Duan, X.; Huang, W.; Gan, C.; Wang, J.; Zhu, W.; and Huang, J. 2018. Weakly supervised dense event captioning in videos. *Advances in Neural Information Processing Systems*, 31.

Ester, M.; Kriegel, H.-P.; Sander, J.; Xu, X.; et al. 1996. A density-based algorithm for discovering clusters in large spatial databases with noise. In *kdd*, volume 96, 226–231.

Feng, C.-M.; Yu, K.; Liu, Y.; Khan, S.; and Zuo, W. 2023. Diverse data augmentation with diffusions for effective test-time prompt tuning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2704–2714.

Gao, J.; Sun, C.; Yang, Z.; and Nevatia, R. 2017. Tall: Temporal activity localization via language query. In *Proceedings of the IEEE international conference on computer vision*, 5267–5275.

Guo, Y.; Liu, J.; Li, M.; Liu, Q.; Chen, X.; and Tang, X. 2024. Trace: Temporal grounding video llm via causal event modeling. *arXiv preprint arXiv:2410.05643*.

Huang, B.; Wang, X.; Chen, H.; Song, Z.; and Zhu, W. 2023. Vtimellm: Empower llm to grasp video moments. *arXiv preprint arXiv:2311.18445*.

Huang, J.; Jin, H.; Gong, S.; and Liu, Y. 2022. Video Activity Localisation with Uncertainties in Temporal Boundary. In *European Conference on Computer Vision*, 724–740. Springer.

Huang, Y.; Yang, L.; and Sato, Y. 2023. Weakly Supervised Temporal Sentence Grounding With Uncertainty-Guided Self-Training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 18908–18918.

Jang, J.; Park, J.; Kim, J.; Kwon, H.; and Sohn, K. 2023a. Knowing where to focus: Event-aware transformer for video grounding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 13846–13856.

Jang, J.; Park, J.; Kim, J.; Kwon, H.; and Sohn, K. 2023b. Knowing where to focus: Event-aware transformer for video grounding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 13846–13856.

Karmanov, A.; Guan, D.; Lu, S.; El Saddik, A.; and Xing, E. 2024. Efficient Test-Time Adaptation of Vision-Language Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 14162–14171.

Kim, D.; Park, J.; Lee, J.; Park, S.; and Sohn, K. 2023. Language-free Training for Zero-shot Video Grounding. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2539–2548.

Krishna, R.; Hata, K.; Ren, F.; Fei-Fei, L.; and Niebles, J. C. 2017. Dense-Captioning Events in Videos. In *2017 IEEE International Conference on Computer Vision (ICCV)*.

Lei, J.; Berg, T. L.; and Bansal, M. 2021. Detecting moments and highlights in videos via natural language queries. *Advances in Neural Information Processing Systems*, 34: 11846–11858.

Li, J.; Li, D.; Savarese, S.; and Hoi, S. 2023. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. *arXiv preprint arXiv:2301.12597*.

Li, J.; Xie, J.; Qian, L.; Zhu, L.; Tang, S.; Wu, F.; Yang, Y.; Zhuang, Y.; and Wang, X. E. 2022. Compositional temporal grounding with structured variational cross-graph correspondence learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 3032–3041.

Li, Z.; Xu, Q.; Zhang, D.; Song, H.; Cai, Y.; Qi, Q.; Zhou, R.; Pan, J.; Li, Z.; Vu, V. T.; et al. 2024. LEGO: Language Enhanced Multi-modal Grounding Model. *arXiv preprint arXiv:2401.06071*.

Luo, D.; Huang, J.; Gong, S.; Jin, H.; and Liu, Y. 2023. Towards Generalisable Video Moment Retrieval: Visual-Dynamic Injection to Image-Text Pre-Training. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 23045–23055.

Luo, D.; Huang, J.; Gong, S.; Jin, H.; and Liu, Y. 2024. Zero-shot video moment retrieval from frozen vision-language models. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 5464–5473.

Ma, H.; Wang, G.; Yu, F.; Jia, Q.; and Ding, S. 2025. MS-DETR: Towards Effective Video Moment Retrieval and Highlight Detection by Joint Motion-Semantic Learning. *arXiv preprint arXiv:2507.12062*.

Mun, J.; Cho, M.; and Han, B. 2020. Local-global video-text interactions for temporal grounding. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 10810–10819.

Nam, J.; Ahn, D.; Kang, D.; Ha, S. J.; and Choi, J. 2021. Zero-shot natural language video localization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 1470–1479.

Qian, L.; Li, J.; Wu, Y.; Ye, Y.; Fei, H.; Chua, T.-S.; Zhuang, Y.; and Tang, S. 2024. Momentor: Advancing video large language model with fine-grained temporal reasoning. *arXiv preprint arXiv:2402.11435*.

Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *ICML*.

Reynolds, D. A.; et al. 2009. Gaussian mixture models. *Encyclopedia of biometrics*, 741(659-663).

Shu, M.; Nie, W.; Huang, D.-A.; Yu, Z.; Goldstein, T.; Anandkumar, A.; and Xiao, C. 2022. Test-Time Prompt Tuning for Zero-Shot Generalization in Vision-Language Models. In Oh, A. H.; Agarwal, A.; Belgrave, D.; and Cho, K., eds., *Advances in Neural Information Processing Systems*.

Wang, G.; Wu, X.; Liu, Z.; and Yan, J. 2022a. Prompt-based Zero-shot Video Moment Retrieval. In *Proceedings of the*

30th ACM International Conference on Multimedia, 413–421.

Wang, Z.; Wang, L.; Wu, T.; Li, T.; and Wu, G. 2022b. Negative Sample Matters: A Renaissance of Metric Learning for Temporal Grounding. In *AAAI*, 2613–2623. AAAI Press.

Wu, J.; Li, G.; Liu, S.; and Lin, L. 2020. Tree-structured policy based progressive reinforcement learning for temporally language grounding in video. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, 12386–12393.

Xu, Y.; Sun, Y.; Xie, Z.; Zhai, B.; and Du, S. 2024a. VTG-GPT: Tuning-Free Zero-Shot Video Temporal Grounding with GPT. *Applied Sciences*, 14(5): 1894.

Xu, Y.; Sun, Y.; Zhai, B.; Jia, Y.; and Du, S. 2024b. Mh-detr: Video moment and highlight detection with cross-modal transformer. In *2024 International Joint Conference on Neural Networks (IJCNN)*, 1–8. IEEE.

Yang, L.; Kong, Q.; Yang, H.-K.; Kehl, W.; Sato, Y.; and Kobori, N. 2023. Deco: Decomposition and reconstruction for compositional temporal grounding via coarse-to-fine contrastive ranking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 23130–23140.

Yang, X.; Feng, F.; Ji, W.; Wang, M.; and Chua, T.-S. 2021. Deconfounded video moment retrieval with causal intervention. In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 1–10.

Yuan, Y.; Lan, X.; Wang, X.; Chen, L.; Wang, Z.; and Zhu, W. 2021. A closer look at temporal sentence grounding in videos: Dataset and metric. In *Proceedings of the 2nd international workshop on human-centric multimedia analysis*, 13–21.

Yuan, Y.; Ma, L.; Wang, J.; Liu, W.; and Zhu, W. 2019. Semantic conditioned dynamic modulation for temporal sentence grounding in videos. *Advances in Neural Information Processing Systems*, 32.

Zeng, X.; Li, K.; Wang, C.; Li, X.; Jiang, T.; Yan, Z.; Li, S.; Shi, Y.; Yue, Z.; Wang, Y.; et al. 2024. Timesuite: Improving mllms for long video understanding via grounded tuning. *arXiv preprint arXiv:2410.19702*.

Zhang, H.; Li, X.; and Bing, L. 2023. Video-llama: An instruction-tuned audio-visual language model for video understanding. *arXiv preprint arXiv:2306.02858*.

Zhang, M. M.; Levine, S.; and Finn, C. 2022. MEMO: Test Time Robustness via Adaptation and Augmentation. In Oh, A. H.; Agarwal, A.; Belgrave, D.; and Cho, K., eds., *Advances in Neural Information Processing Systems*.

Zhang, R.; Zhang, W.; Fang, R.; Gao, P.; Li, K.; Dai, J.; Qiao, Y.; and Li, H. 2022. Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification. In *European Conference on Computer Vision*, 493–510. Springer.

Zhang, S.; Peng, H.; Fu, J.; and Luo, J. 2020. Learning 2d temporal adjacent networks for moment localization with natural language. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, 12870–12877.

Zheng, M.; Cai, X.; Chen, Q.; Peng, Y.; and Liu, Y. 2024. Training-free video temporal grounding using large-scale pre-trained models. In *European Conference on Computer Vision*, 20–37. Springer.

Zheng, M.; Gong, S.; Jin, H.; Peng, Y.; and Liu, Y. 2023. Generating Structured Pseudo Labels for Noise-resistant Zero-shot Video Sentence Localization. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 14197–14209.

Zheng, M.; Huang, Y.; Chen, Q.; and Liu, Y. 2022a. Weakly Supervised Video Moment Localization with Contrastive Negative Sample Mining. In *Proceedings of the AAAI Conference on Artificial Intelligence*.

Zheng, M.; Huang, Y.; Chen, Q.; Peng, Y.; and Liu, Y. 2022b. Weakly Supervised Temporal Sentence Grounding with Gaussian-based Contrastive Proposal Learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*.

Reproducibility Checklist

Instructions for Authors:

This document outlines key aspects for assessing reproducibility. Please provide your input by editing this `.tex` file directly.

For each question (that applies), replace the "Type your response here" text with your answer.

Example: If a question appears as

```
\question{Proofs of all novel claims
are included} {(yes/partial/no)}
Type your response here
```

you would change it to:

```
\question{Proofs of all novel claims
are included} {(yes/partial/no)}
yes
```

Please make sure to:

- Replace ONLY the "Type your response here" text and nothing else.
- Use one of the options listed for that question (e.g., **yes**, **no**, **partial**, or **NA**).
- **Not** modify any other part of the `\question` command or any other lines in this document.

You can `\input` this .tex file right before `\end{document}` of your main file or compile it as a stand-alone document. Check the instructions on your conference's website to see if you will be asked to provide this checklist with your paper or separately.

---

1. General Paper Structure

1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) yes

1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) yes

1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) yes

2. Theoretical Contributions

2.1. Does this paper make theoretical contributions? (yes/no) yes

If yes, please address the following points:

2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) partial

2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) partial

2.4. Proofs of all novel claims are included (yes/partial/no) partial

2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) partial

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) partial

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) partial

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) partial

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no) yes

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) partial

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) NA

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) yes

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) NA

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) yes

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA) NA

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no) yes

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) partial

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) no

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) partial

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) partial

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) partial

4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) NA

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) partial

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) partial

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) no

4.11. Analysis of experiments goes beyond single-

dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) yes

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) partial

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) partials